

Jeonggyo Yoon

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Education

Ph.D. Candidate in Environmental Health Sciences

Department of Community Environment & Policy, Mel & Enid Zuckerman College of Public Health,
University of Arizona

Aug. 2021- Present

Advisor: Dr. Chris Lim

Dissertation: "Air pollution, greenness, and cancer-related outcomes"

M.S in Humanities and Social Medicine

Department of Medicine, Graduate School, Gachon University, South Korea

Mar. 2017-Feb. 2019

Advisor: Dr. Yoon-Hyeong Choi

Master's Thesis: "Association between phthalate exposures and walking speed in older Korean adults"

B.E in Biomedical Engineering

Gachon University, College of Health Science, South Korea

Mar. 2012-Feb. 2017

Research Experience

Graduate Research Associate

University of Arizona, MEZCOPH, Department of Community, Environment & Policy

Aug. 2021-Present

- **Air Pollution Meta-analysis** (manuscript in preparation):
 - Screened more than 12,000 records, categorized approximately 200 studies for systematic review and meta-analysis.
 - Extracted study-level data and conducted quantitative synthesis of air pollution and health effects across multiple exposure and health outcomes.
 - Developed visual summaries and figures to facilitate interpretation of pooled findings.
- **Harmful Algal Blooms and Health:**
 - Reviewed and synthesized evidence from 30+ studies on harmful algal blooms and health effects.
 - Conducted literature investigation, curated relevant resources, and contributed to manuscript writing and revision, helping identify key evidence gaps in environmental health impacts.
- **Climate Hazards, Infrastructure, and Equity** (manuscript in preparation):
 - Conducted geospatial and equity analyses linking 4.2 million U.S. infrastructure facilities across 84,093 census tracts to 15 climate hazards
 - Developed consequence-based classifications for 9 facility types and identified 2,971 triple-risk tracts where hazard exposure, social vulnerability, and low resilience converge.
 - Generated evidence to inform equity-focused climate adaptation and infrastructure planning.
- **Neighborhood Environment and Cancer:**
 - Examined geocoded residential exposures to air pollution and greenspace in relation to obesity-related cancer outcomes in the NIH-AARP Diet and Health Study, a prospective cohort of more than 567,000 participants.
 - Designed and implemented an analytical framework for time-varying and lagged exposure modeling, confounder adjustment, effect modification, and mediation analyses.

Gachon University College of Medicine, Department of Preventive Medicine

Mar. 2020-Aug.2021

- Performed ecological study to provide epidemiological evidence that HD (Humidifier Disinfectant) exposure is associated with the development of severe lung injury cases in Korea.
- Investigated changes in the nationwide number of humidifier disinfectant-associated lung injury (HDLI) in concordance with nationwide HD consumption in South Korea using data covering the entire Korean population.

Researcher

Environmental Health and Epidemiology Laboratory (EHEL), Gachon University

Mar. 2019-Feb. 2020

- Planned and organized the whole process of research project (Body Burden Project: Reducing Endocrine Disrupting Chemicals (EDCs) in Daily Life): Prepared IRB approval, developed questionnaires and codebook for obtaining study information, adjusted examination schedule, and handled data management.
- Conducted on-site investigation in large hospitals (Investigation of Using Humidifier Disinfectant in Public Use Facilities) and interviewed nurses and administrative assistants to understand background information on the HD exposure environment at hospitals.
- Gathered, arranged and corrected data to create representative graphs and tables highlighting results for research publication.

Postgraduate Research Assistant

Gachon University College of Medicine, Department of Preventive Medicine

Mar. 2017-Feb.2019

- Explored the effects of environmental phthalate exposure on physical function (i.e., walking speed, weight loss, frailty) in Korean elderly population.
- Analyzed data using SAS and performed mixed effect models to evaluate the association between phthalate exposure and walking speed from repeated measures data.

Undergraduate Intern

Gachon University College of Medicine, Department of Preventive Medicine

July 2016 – Feb. 2017

- Investigated the relationship of sarcopenia with rheumatoid arthritis and osteoarthritis using KNHANES data.
- Analyzed data using SPSS and utilized logistic regression models to evaluate the relationships while controlling for major risk factors.
- Interpreted statistical results for presentation and worked with supervisor to prepare for domestic publication.

Manager (Cycle IV-2) / Field Interviewer (Cycle IV-1/2/3)

Korean Society of Environmental Health (KSEH)

Nov. 2016 – Mar. 2019

Korea Environment Industry and Technology Institute (KEITI), “Investigation on Humidifier Disinfectant-associated Lung Injury (Cycle IV)”

- Trained and managed field interviewers affiliated with over 20 local university investigation centers in Korea.
- Coordinated multi-center implementation of a national humidifier disinfectant exposure investigation involving more than 4,500 applicants, including workload distribution, data quality review, and follow-up on incomplete submissions.

- Conducted epidemiologic field investigations on HD exposure cases: Corresponded promptly with all interviewees, coordinated schedule and conducted interviews.

Publications & Presentations

Publications

1. M Hong, MJ Ju, **J Yoon**, W Lee, S Lee, E Jo, S Choi, W Yang, Y Choi. Exposures to humidifier disinfectant and various health conditions in Korean based on personal exposure assessment data of claimants for compensation. *BMC Public Health*, 23(1), 1800. October 2023.
<https://doi.org/10.1186/s12889-023-16389-x>
2. CC Lim, **J Yoon**, LB Gerald, AB Ault, S Heo, ML Bell. *Harmful algal bloom aerosols, climate change, and human health: A review*. *EBioMedicine*. May 2023. <https://doi.org/10.1016/j.ebiom.2023.104604>
3. **J Yoon**, E García -Esquinas, J Kim, JH Kwak, H Kim, S Kim, K Kim, Y Hong, Y Choi. *Urinary phthalate metabolites and slow walking speed in the Korean Elderly Environmental Panel II study*. *Environmental Health Perspectives*. April 2023. <https://doi.org/10.1289/EHP10549>
4. S Lee, K Han, **J Yoon**, E Jo, W Yang, Y Choi. *New-Onset and Exacerbation of Lung Diseases after Short-Term Exposures to Humidifier Disinfectant during Hospitalization*. *Toxics*. 2022; 10(7):371.
<https://doi.org/10.3390/toxics10070371>
5. MJ Ju, **J Yoon**, E Jo, S Lee, J Oh, J Park, W Yang, Y Choi. *A Study on the Regional Distribution of Enrollment Cases Who Reported Humidifier Disinfectant-Related Health Effects*. *Journal of Environmental Health Sciences*. Aug 2021;47(4):303-9 (in Korean).
<https://doi.org/10.5668/JEHS.2021.47.4.303>
6. H Ryu, Y Choi, E Kim, J Park, S Lee, **J Yoon**, E Jo, Y Choe, J Heo, W Yang. *Misclassification and characterization of exposure to humidifier disinfectants using a questionnaire*. *BMC public health*. Jul 2021;21(1):1-10. <https://doi.org/10.1186/s12889-021-11459-4>.
7. A Seong, S Heo, **J Yoon**, J Lee, M Choi, J Jeon, J Kim, Kwi Park, J Im, K Ko, J Jung, Y Choi. *Academic Stress and Its Influencing Factors among Medical and Non-Medical Students*. *Korean Medical Education Review*. Jun 2021, 23(2), 128-138 (in Korean).
<https://doi.org/10.17496/kmer.2021.23.2.128>.
8. **J Yoon**, M Kang, J Jung, MJ Ju, SH Jeong, W Yang, Y Choi. *Humidifier disinfectant consumption and humidifier disinfectant-associated lung injury in South Korea: A nationwide population-based study*, *International journal of environmental research and public health*. Jun 2021, 18(11), 6136.
<https://doi.org/10.3390/ijerph18116136>.
9. K Han, **J Yoon**, E Jo, H Ryu, W Yang, Y Choi, *Case Studies of exposures to humidifier disinfectant in hospitals: Focusing on the exposure assessment of the fourth round of applicants*, *Journal of Environmental Health Sciences*. Aug 2019, 45(4): 358-369 (in Korean).
<https://doi.org/10.5668/JEHS.2019.45.4.358>. **(The Best Science and Technology Paper Award)**
10. S Lee, **J Yoon**, J Ock, E Jo, H Ryu, W Yang, Y Choi, *Individual exposure characteristics to humidifier disinfectant according to exposure classification groups –Focusing on 4-1 and 4-2 applicants–*, *Journal of Environmental Health Sciences*. Aug 2019, 45(4): 370-380 (in Korean).
<https://doi.org/10.5668/JEHS.2019.45.4.370>
11. H Ryu, E Jo, Y Choi, S Lee, **J Yoon**, JH Kwak, J Park, J Heo, P Kim, W Yang, *Analysis of affecting factors on exposure assessment errors and characteristics of applicants for damage by usage of humidifier disinfectants*, *Journal of Environmental Health Sciences*. Feb 2019, 45(1): 71-81 (in Korean).
<https://doi.org/10.5668/JEHS.2019.45.1.71>

12. Y Choi, H Ryu, **J Yoon**, S Lee, JH Kwak, B Han, Y Chu, P Kim, W Yang, *Demographic characteristics and exposure assessment for applicants who have been injured by humidifier disinfectant –Focusing on 4-1 and 4-2 applicants–*, Journal of Environmental Health Sciences. Aug 2018, 44(4): 301-314 (in Korean). <https://doi.org/10.5668/JEHS.2018.44.4.301>
13. **J Yoon**, J Kim, Y Choi, *Associations of sarcopenia with rheumatoid arthritis and osteoarthritis in Korean adults: The Korea National Health and Nutrition Examination Survey 2008-2011*, The Journal of Kinesiology. Jul 2017, 19(3): 61-70 (in Korean). DOI: <https://doi.org/10.15758/jkak.2017.19.3.61>. (**Best Paper Award**)

Presentations

1. **J Yoon**, R Jones, J Fisher, G Thurston, C Lim. *Associations between air pollution and cancer survivorship in the U.S: the NIH-AARP Diet and Health study*. Poster Presentation at the ISES (International Society of Exposure Science)/ ISEE (International Society for Environmental Epidemiology) joint conference, Atlanta, GA (2025.08.18)
2. **J Yoon**, C Lim. *Natural hazards and critical infrastructure in the United States*. Oral Presentation at the ISES (International Society of Exposure Science), Montreal, Canada (2024.10.23)
3. **J Yoon**, C Lim, G Thurston. *Associations between Fine Particulate Matter Composition and Sources with Adverse Health Effects: A Systematic Review and Meta-Analysis*. Poster Presentation at the 35th Annual Conference of ISEE (International Society for Environmental Epidemiology), Kaohsiung, Taiwan (2023.09.19)
4. **J Yoon**, C Lim. *Natural hazards and public health infrastructure in the United States*. Poster Presentation at the 35th Annual Conference of ISEE (International Society for Environmental Epidemiology), Kaohsiung, Taiwan (2023.09.19)
5. **J Yoon**, C Lim. *Natural hazards and public health infrastructure in the United States*. Poster Presentation at MEZCOPH Public Health Poster Forum, AZ, US (2023.03.31) (**1st place of Research Awards**)
6. **J Yoon**, M Kang, J Jung, MJ Ju, SH Jeong, W Yang, Y Choi. *Humidifier disinfectant sales and humidifier disinfectant-associated lung disease in South Korea: A National Health Insurance Service database*. Poster presentation at Fall Conference of Korean Society of Environmental Health, Online conference, Korea (2020.11.13)
7. **J Yoon**, J Kim, JH Kwak, K Kim, H Kim, S Kim, Y Hong, Y Choi. *Associations between phthalate exposures and walking speed in Korean elderly*. Oral Presentation at Fall Conference of Korean Society of Environmental Health, Seoul, Korea (2018.11.15) (**Best Graduate Student Oral Presentation Award**)
8. **J Yoon**, J Kim, JH Kwak, K Kim, H Kim, S Kim, Y Hong, Y Choi. *Associations between urinary phthalate metabolites and walking speed in elderly adults*. Poster presentation at Spring Conference of Korean Society of Environmental Health and Toxicology, Seoul, Korea, (2018.10.19) (**Best Poster Presentation Award**)
9. **J Yoon**, J Kim, JH Kwak, K Kim, H Kim, S Kim, Y Hong, Y Choi. *Association between phthalate exposure and frailty in Korean older adults*. Poster presentation at Spring Conference of Korean Society of Environmental Health and Toxicology, Seoul, Korea, (2018. 06.01)
10. **J Yoon**, MJ Ju, SG Park, JH Kwak, Y Choi. *Case reports of exposures to humidifier disinfectant in Incheon: Focusing on the exposure assessment of the fourth round of applicants*. Oral Presentation at

Spring Conference of Korean Society of Environmental Health, Seoul, Korea (2018.05.31) (**Best Professional Field Interviewer Award**)

11. **J Yoon**, J Kim, Y Choi. *Associations of sarcopenia with rheumatoid arthritis and osteoarthritis in middle-aged and older Korean adults*. Poster presentation at Fall Conference of Korean Society for Preventive Medicine, Busan, Korea (2017. 10. 20) (**Best Poster Presentation Award**)

Teaching, Mentoring, and Outreach

Graduate Teaching Assistant

University of Arizona, MEZCOPH, Department of Community, Environment & Policy Aug. 2023-Jan.2024

- Supported instruction for *EHS 575: Environmental and Occupational Health*, an online graduate-level course with 69 students.
- Assisted with course management on D2L, responded to student questions, and provided academic support via Zoom appointments.
- Supported assessment of discussions, quizzes, assignments, and term paper components related to environmental hazard assessment, risk management, and environmental justice.

University of Arizona, MEZCOPH, Department of Health Promotion Sciences Aug. 2022-Jan.2023

- Supported instruction for *HPS 300: Public Health in the 21st Century*, an undergraduate course with 56 students.
- Provided student support and feedback on public health writing, discussion activities, assignments, and presentations focused on health disparities, climate change, and other contemporary public health issues.

Mentoring and Outreach

Research Mentor, *STEPS 2 STEM, University of Arizona* June.2025

- Mentored high school students in environmental health research through structured activities including lab tours, paper discussions, data extraction training, and student project development.
- Guided students in interpreting scientific literature on air pollution and health, reviewing study design and results, and preparing a final student exhibit.

Guest Lecturer, *Patagonia Union High School, Patagonia, AZ* May.2025

- Presented to 11th- and 12th-grade science students on air pollution and health risks through a community-engaged air quality monitoring context, emphasizing how local data can support environmental health awareness.

Honors & Awards

Scholarships and Travel Support

CRTEC Travel Award, Cancer Research Training and Education Coordination, University of Arizona.	2025
GPSC Travel Award, Graduate and Professional Student Council, University of Arizona.	2025
Dean's Annual Fund Scholarship, College of Public Health, University of Arizona.	2023
CRTEC Travel Award, Cancer Research Training and Education Coordination, University of Arizona.	2023

Research and Presentation Awards

Best Poster Presentation Award, MEZCOPH Public Health Poster Forum, University of Arizona.	2023
Co-author of award-winning paper, 30 th The Best Science and Technology Paper Awards,	2020

The Korean Federation of Science and Technology Societies (KOSFT)	
Best Graduate Student Oral Presentation Award, The Korean Society of Environmental Health	2018
Best Poster Presentation Award, The Korean Society of Environmental Health and Toxicology	2018
Best Professional Field Interviewer Award, The Korean Society of Environmental Health	2018
Best Paper Award, The Journal of Kinesiology	2018
Best Poster Presentation Award, The Korean Society for Preventive Medicine	2017

Professional Memberships

International Society for Environmental Epidemiology (ISEE)
 International Society of Exposure Science (ISES)
 Society for Epidemiologic Research (SER)
 Korean-American Scientists and Engineers Association (KSEA)

Skills

Statistical Analysis: Cox proportional hazards, mixed-effects models, logistic regression, BKMR, Machine learning, meta-analysis, geospatial analysis, spatial autocorrelation, repeated-measures analysis

Proficient programs: SAS, SPSS, R, Stata

Soft Skills: Communication, strong work ethic, teamwork, conflict resolution, problem-solving